SMART INDIA HACKATHON 2024

TITLE PAGE

- **Problem Statement ID – 1648**
- **Problem Statement Title – Online Chatbot based ticketing system.**
- **Theme – Travel & Tourism**
- **PS Category – Software**
- **Team ID – NaN**
- **Team Name – Code Monkeys**

PROPOSED SOLUTION

SOLUTION →

Development of a **Multilingual AI Chatbot** based **Ticketing System** specialized for diverse linguistic communities.

- **Language Detection:** Automatically identifies and switches to the user's preferred language.
- **User-Friendly Interface**: Simplifies the ticket booking process with easy to use interface.
- **Secure Payment Gateway**: Integrates a payment gateway that supports multiple payment methods, ensuring convenient transaction process.
- **Automated Email Confirmation**: Sends an email with transaction details, tickets and event information in the user's preferred language.
- **No Form Filling Process:** Chatbot will fill booking form for the users in backend.

UNIQUENESS →

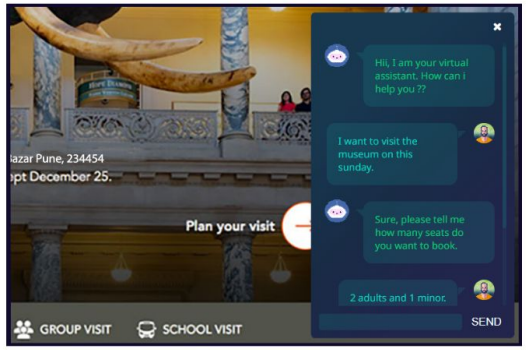
- Use of free and open source AI Model i.e **Meta LLama**.
- **Real Time Analytics**:Track user behavior and trends in real-time, providing actionable data to optimize services and enhance user engagement.
- …

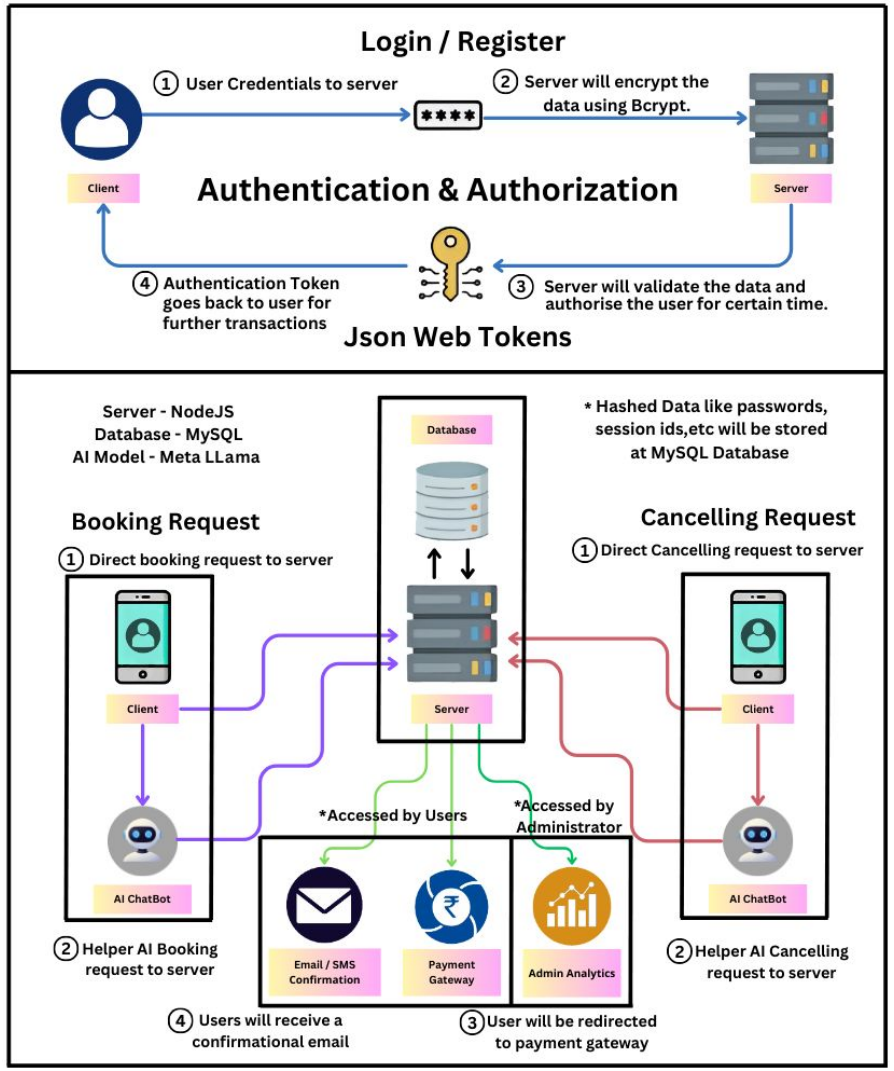
ADDRESSING THE PROBLEM →

Our **multilingual** AI chatbot transforms ticketing, allowing users to buy museum tickets in their language with real-time, culturally tailored interactions that adapt to national diversity.

TECHNICAL APPROACH

- **Backend** : Core technologies are **Node JS** and **Express JS**, **MySQL** database.
- **Frontend** : **React JS** and **Tailwind** will be used to make beautiful User Interface for the website.
- **Cloud** : Google Cloud Platform
- **Payment Gateway** : **Razorpay** with upi, Credit/Debit Card, Netbanking etc.
- **APIs** : Google Translation, Google Analytics.

Tech Stack			Development in Progress
Front-End	Back-End	Others	
React JS	Node JS	SQL Database	
Tailwind	Express JS	Razorpay UPI Gateway	
Bootstrap	Python for AI	Google Analytics	



FEASIBILITY AND VIABILITY

Analysis of Feasibility →

- Our multilingual AI chatbot-based ticketing system is not only easy to build and operate but also financially smart and impactful.
- The tech stack can expand efficiently by adding resources or more servers to handle more users and data.
- The cost of hosting the web application:-
  - Google Cloud E2 Medium Instance - **₹15,000** per month.
  - Cloud SQL Database - **₹8,000** per month.
  - AI model (META LLAMA) - **₹0** because AI model will run locally, hence no additional cost.

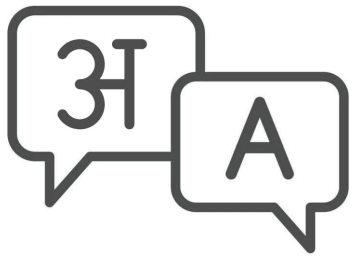
Potential Risks and Challenges →

- **Challenge**: Adding multi-language compatibility to the Large Language Model (AI Chatbot).
- **Mitigation**:Google Translation API can be used instead of training Large Language Model with multiple-languages.

Cost Comparision

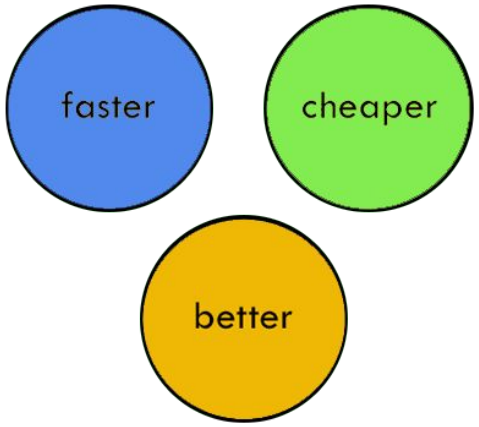
ChatGPT	+	AWS	=	₹80,000 / mo
Gemini	+	Google Cloud	=	₹70,000 / mo
LLama	+	Google Cloud	=	₹25,000 / mo
Vertex AI	+	AWS	=	₹55,000 / mo

*More details on research page



IMPACT AND BENEFITS

- This chatbot system could lead to a **80%** reduction in ticketing errors, such as double bookings or lost records.
- Chatbots can reduce booking time by **75%**, with an average booking time dropping from **10 minutes** to **2.5 minutes**, especially during peak hours.
- Museums implementing chatbot systems can save up to **50%** on **operational costs** by minimizing the need for staff dedicated to ticketing, customer queries, and queue management.
- By accommodating the **90%** of Indian internet users who prefer services in their local language, a **multilingual AI** chatbot makes it simple for people from all over India to buy tickets.
- The chatbot can implement **dynamic pricing** based on factors like time of day or special events, along with **membership programs** for frequent visitors, driving loyalty and offering exclusive benefits and discounts.



RESEARCH AND REFERENCES

- **LLama research →**
  - The Meta AI team's extensive research on LLaMA's language capabilities.
  - https://ai.meta.com/research/publications/the-llama-3-herd-of-models/
- **AI training for chatbots →**
  - https://www.museumnext.com/article/how-are-museums-are-using-chatbots/
  - Dataset → https://github.com/allenai/OpenBookQA
- **List of chatbots being used in museums around the world →**
  - https://zkm.de/en/talk-to-me-chatbots-in-museums#:~:text=Also%20customized%20chatbots%2C%20some%20programmed,with%20historical%20objects%20and%20places.
- **APIs Cost Comparison →**
  - https://drive.google.com/file/d/1LLBtgHimlXVUyjpfESrEeoc6QCTW6kZm/view?usp=drive_link
- **Payment Gateway Documentation →**
  - https://razorpay.com/docs/payments/payment-gateway

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- **Real Time Analytics** .....
- 2324234234 sdfs sf ……

ADDRESSING THE PROBLEM →

Our multilingual AI chatbot transforms ticketing by allowing users to buy museum tickets in their preferred language, offering real-time, culturally-tuned interactions that make navigating diverse global nuances effortlessly engaging.

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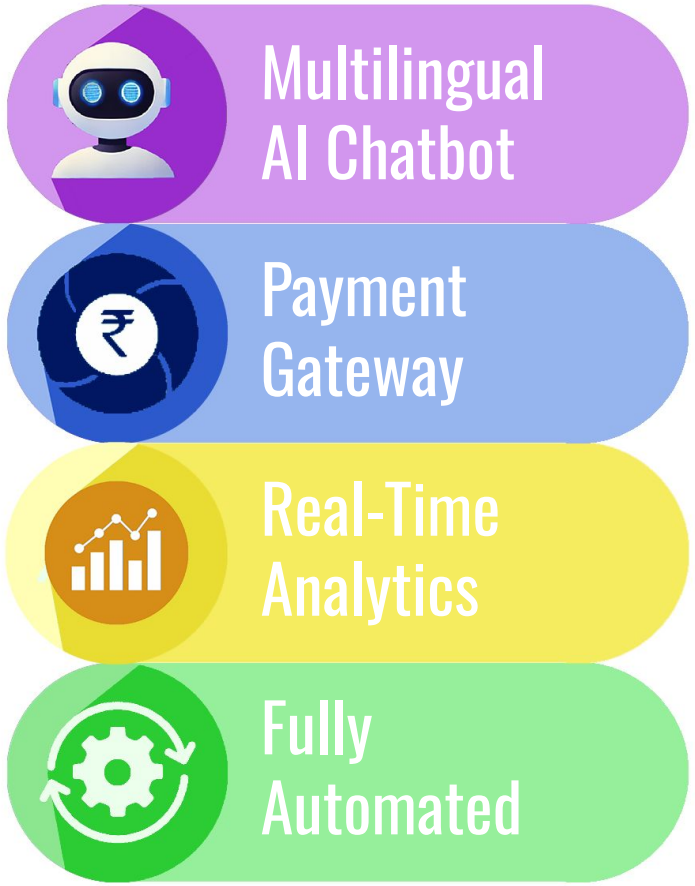
PROPOSED SOLUTION

- **Our solution features a full stack web application hosted on google cloud platform. It features →**

- **Meta LLAMA** based multilingual **AI chatbot** that facilitates ticket booking, integrated with a secure online payment gateway via **Razorpay**.
- Real-time data **analytics** for tracking visitor trends and provides a user-friendly interface built with **React JS** and **Tailwind CSS**.
- This automated system ensures a smooth process for ticket issuance and booking confirmations through **Email** and **SMS**, accessible across all devices.
- The system offers **No Form Filling** process, Chatbot will fill booking form for the users in backend, making the whole booking experience better.